Multi-Objective Dynamic Startup Scheduling Technique Based on Improved Nondominated Sorting Genetic Algorithm II

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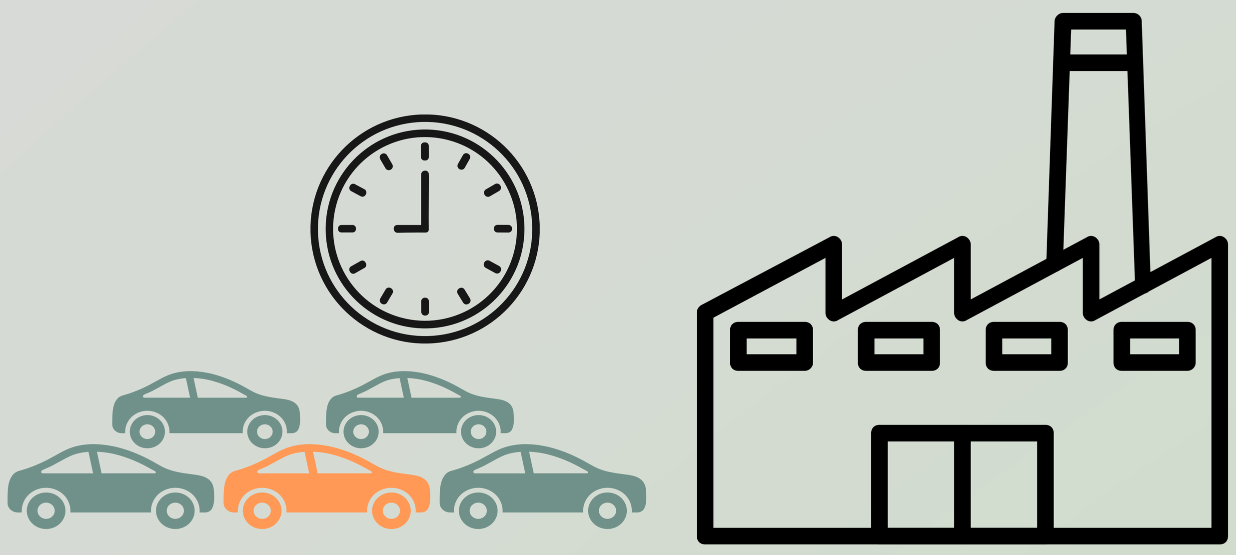
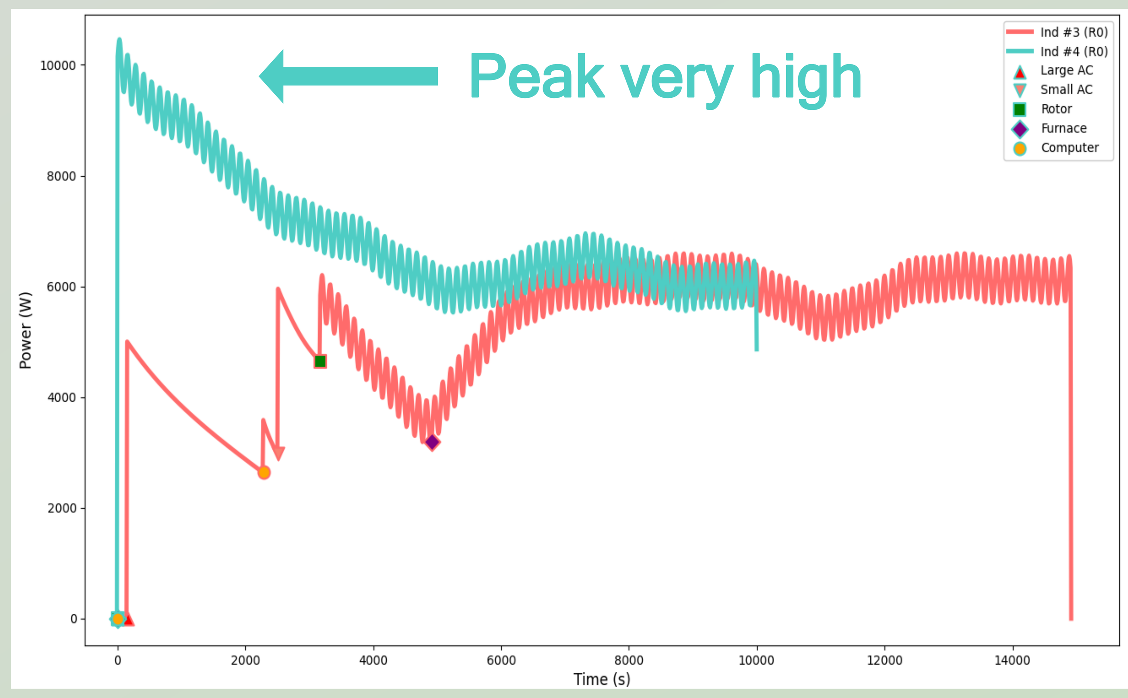
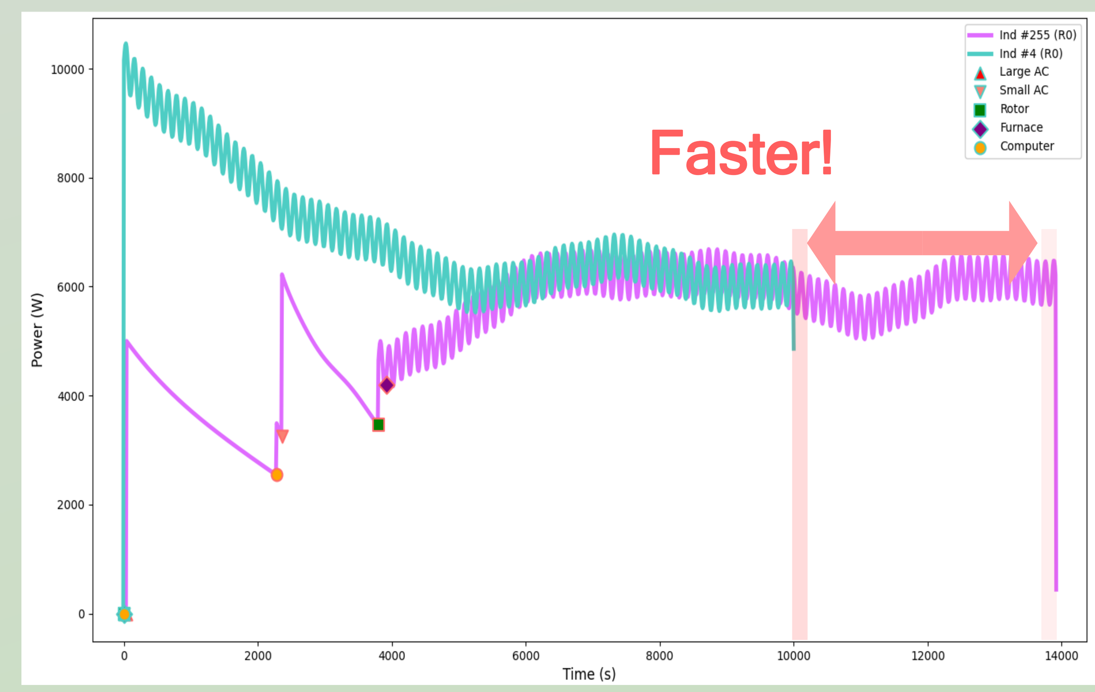
Introduction

Main Contribution : To provide an optimization algorithm that achieves efficient, stable, and low-peak machine startup schedules.



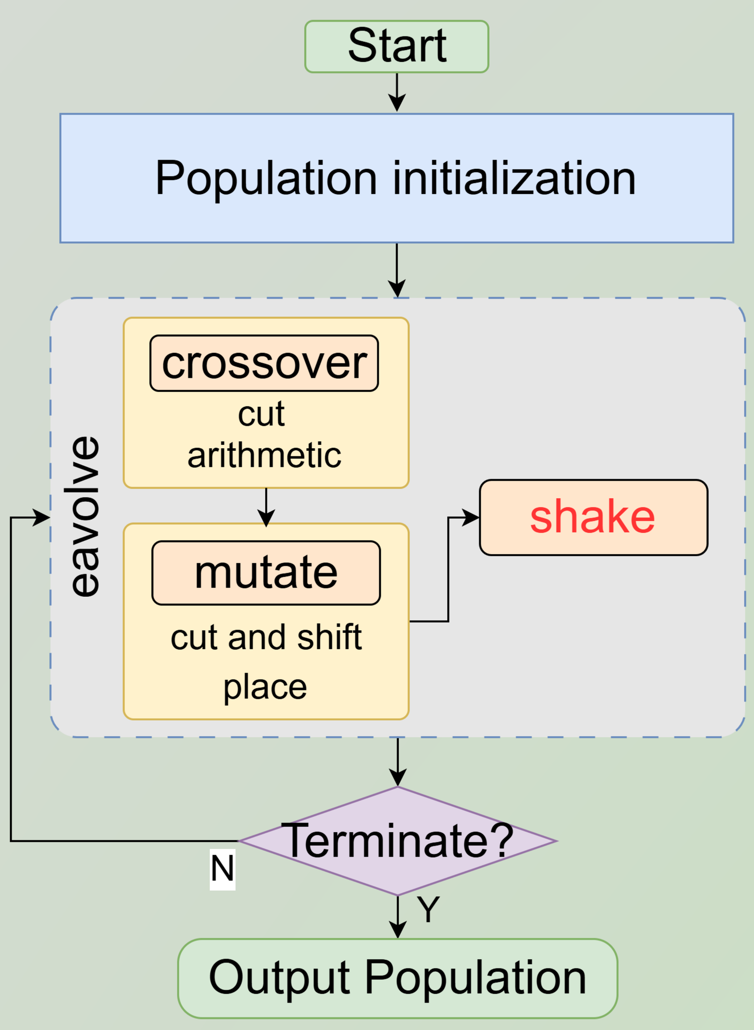


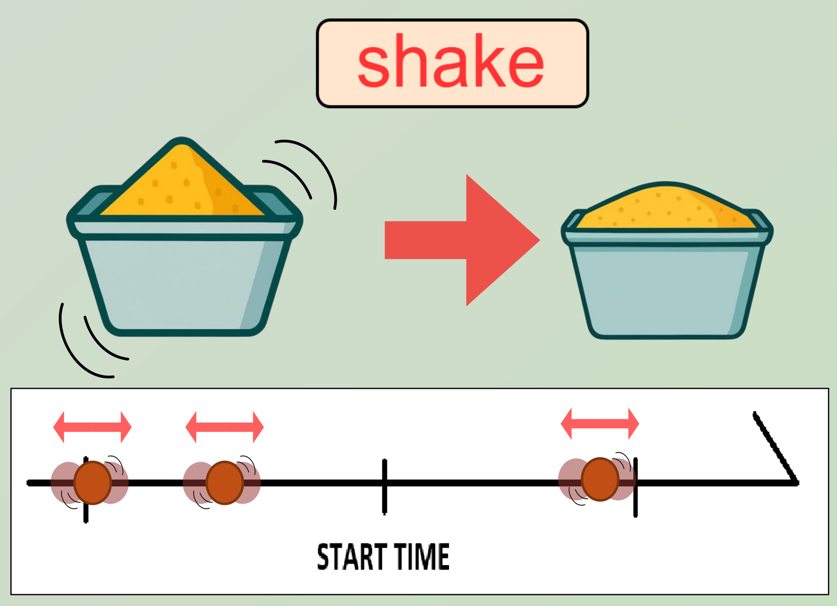


For Peak

For Time

The Proposed Approach

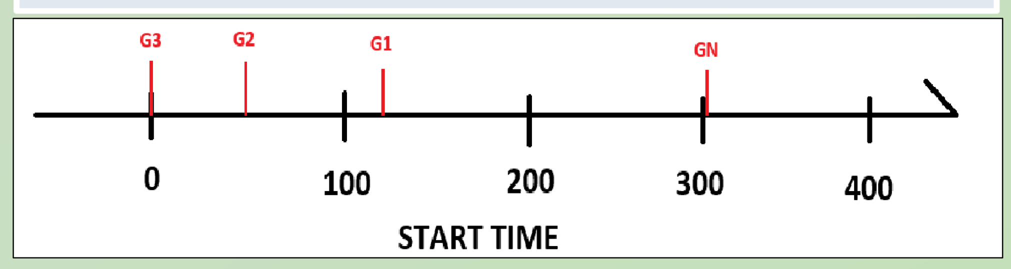




Chromosome

Gene 1	Gene 2	Gene 3	...	Gene N
120.5	45.2	0.0	...	300.1

Values represent: Startup Delay Time



SET SLOWEST(slowest machine time)=1000
Step=10,N(machine count)=100

Solution Space $=(\frac{SLOWEST}{Step})^N = O(100^N)$

$this\ case: 100^{100}$

The total atoms in the observable universe is 10^{80}

Fitness Functions

Obj1:Peak Power

f_p=Max(Power at any time)

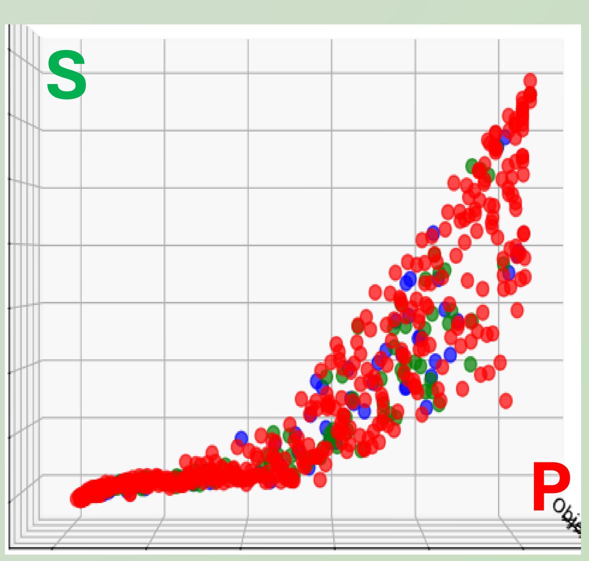
Obj2:Finish Time

$f_T = \sum(genes)$ +10*max(genes)

Obj3:Power Smooth

$f_S = Avg((P_t - P_{t-1})^2)$

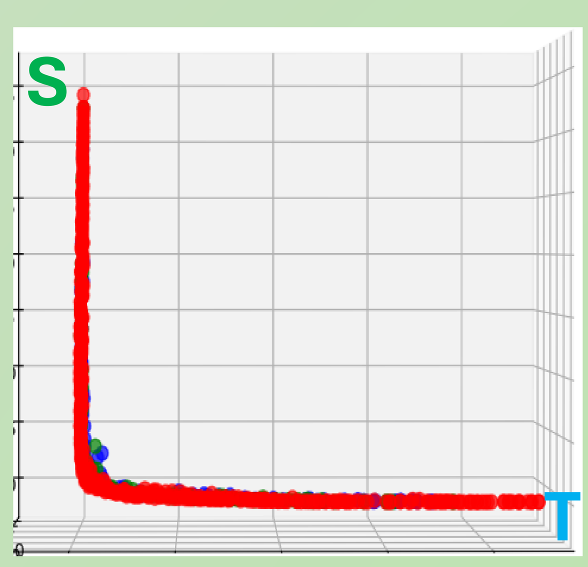
Result (Generations:500 Group:400 Gene_size:100)

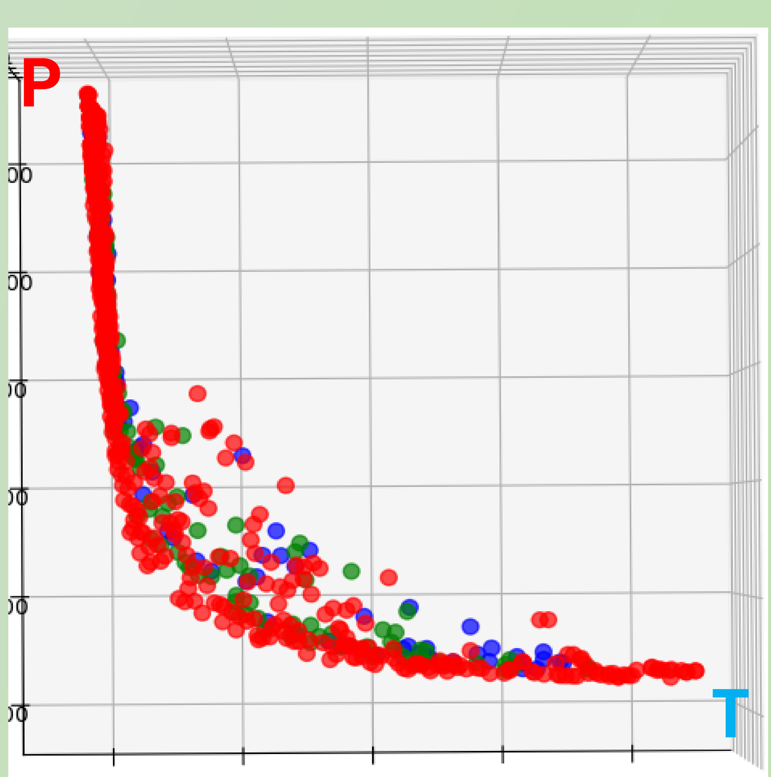


lower is batter

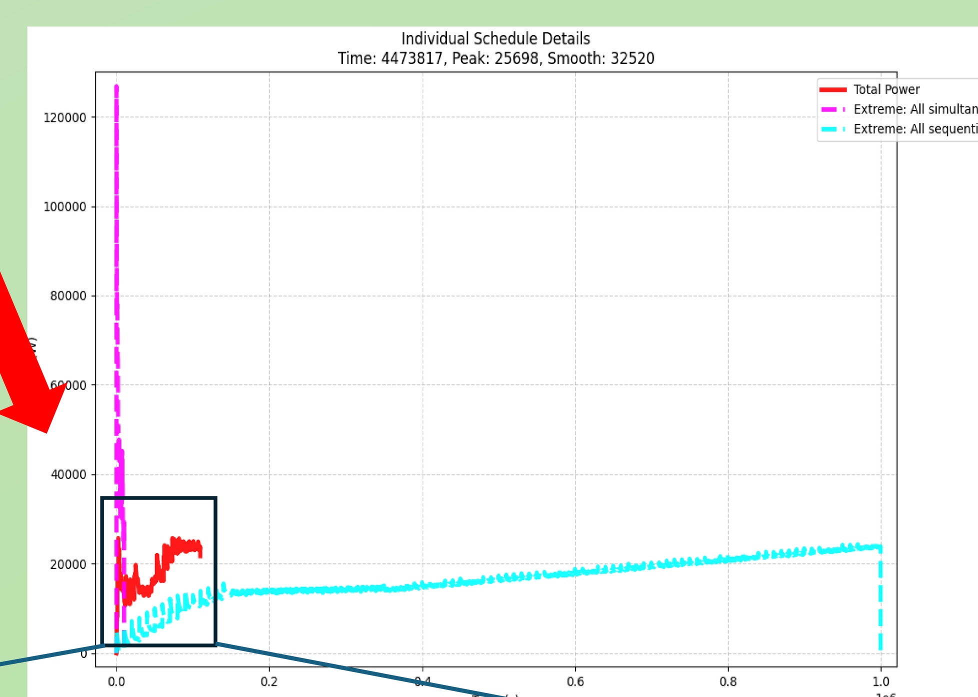
Not conflict

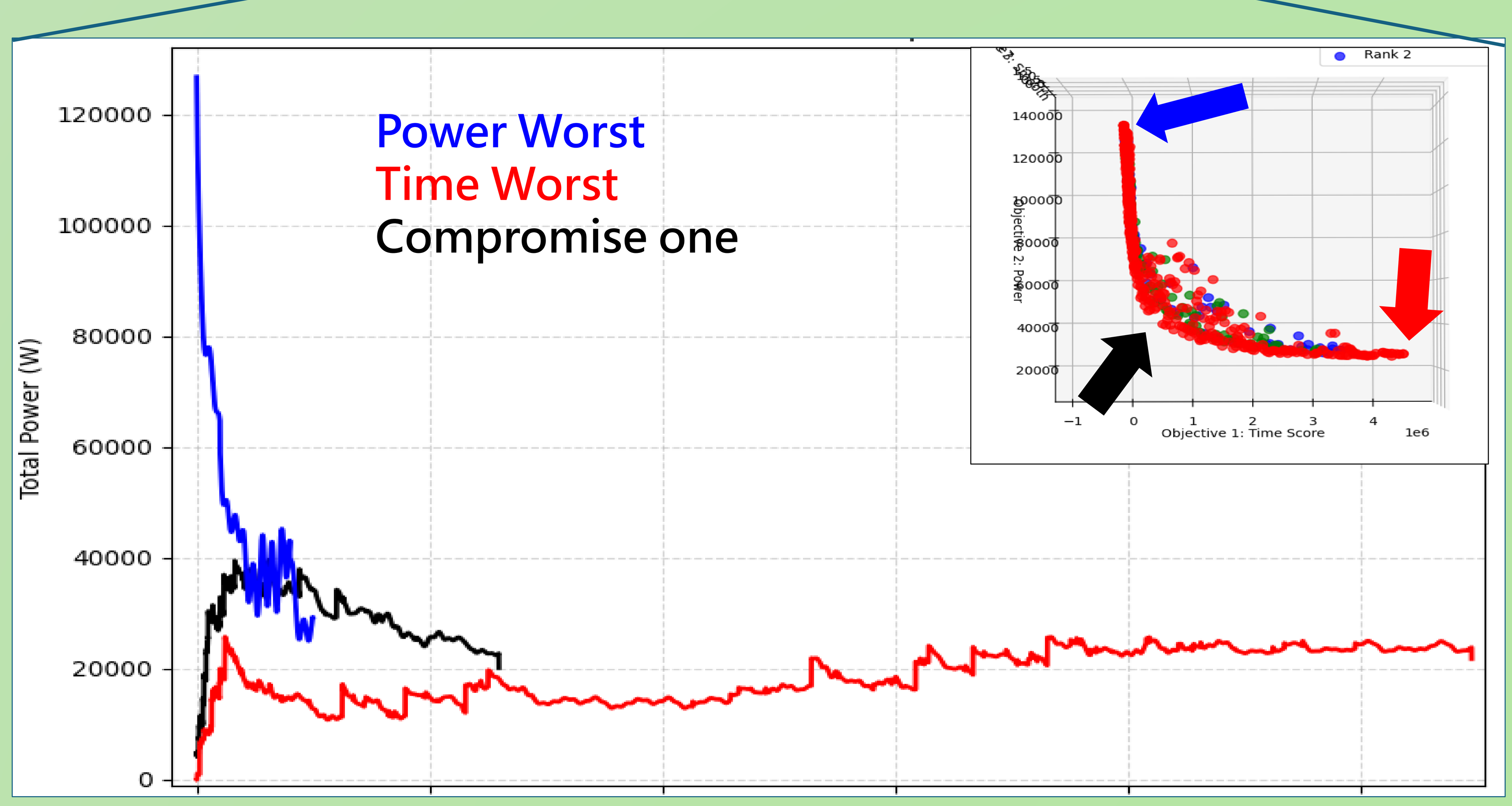
Trade-off





Time Worst Individual Also not bad





3D S-P-T view

Conclusions

- improve NSGA-II combined with a novel "shake"
- Dynamic and continuous analog power consumption integration
- Provide usable power consumption sorting methods

This project was supported by the Center for Teaching and Learning Development, Office Academic Affairs, National Kaohsiung University of Science and Technology under the grant named: Teaching Enhancement Program.